

Research on Charge Accumulation Characteristics by PEA Method and $Q(t)$ Method

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Abstract—This article reports the evaluation system of the insulating properties based on the pulsed electro-acoustic (PEA) method and the current integration charge $Q(t)$ method. According to the basic principles of the methods, the relationship between the charge ratio and dielectric relaxation time in $Q(t)$ data is first derived, by which a new method is proposed to calculate the material conductivity. By further analyzing the charge results of different materials, it is pointed out that the PEA method focuses on the related characteristics of the net charge distribution captured by deep traps, while the $Q(t)$ method reflects the changing process of the total charge amount and related characteristics. No single method can completely analyze all the charge characteristics of the material, and thus the necessity of the combination of the two methods is determined. Meanwhile, based on the theoretical derivation and experimental results, it is indicated that when the charge ratio of $Q(t)$ data exceeds 2, the electrical state inside the sample basically stabilizes. At this stage, the distorted electric field can be analyzed by the PEA method, and the $Q(t)$ method can calculate the leakage current, dielectric relaxation time, and conductivity. Therefore, the combination of the two methods can provide a more reasonable assessment for the insulation property and state.

Index Terms—Current integration charge method, electric charge accumulation, insulating property, leakage current, pulsed electro-acoustic (PEA) method.

I. INTRODUCTION

THE insulation of HVDC equipment usually operates under the joint effect of high temperature and high electric field, which thus suffers from more serious problems such as space charge accumulation and leakage current than traditional ac equipment [1]–[3]. These electrical phenomena can further cause electric field distortion, electric tree development, and partial discharge, which can easily lead to

degradation of insulating materials [4]–[6]. Therefore, it is necessary to establish an accurate technology for evaluating HVDC insulation. At present, the measurement methods of the insulation state mainly contain the picoammeter method to measure the current state, the methods to present space charge distribution, and the current integration charge $Q(t)$ method. The accurate evaluation of the insulation property relies on the reliability of these measurement techniques.

Among the methods, the results of picoammeter can reflect the changing process of dielectric current. However, it is often affected by a lot of random noise, and thus the method has limitations in some application environments [7], [8]. The methods for measuring space charge distribution mainly contain the pulsed electro-acoustic (PEA) method, pressure wave methods, and thermal methods [9]–[11]. Due to the advantages of the simple principle and easy construction, the PEA method proposed by Takada has become one of the widely used measurement methods [12]–[14]. It can effectively obtain the charge distribution captured by deep traps inside insulating films and treated cables. However, it has limitations in the measurement situation of shallow trapped charges and actual power equipment [15], [16]. In response to this problem, Takada further proposed and developed the $Q(t)$ method [17]. It can obtain the whole charge information including the induced surface charge, the internal space charge, and the leakage charge of the insulation. Meanwhile, similar to the research on deep and shallow trapped charges based on dielectric responses [18], the $Q(t)$ method is also dedicated to noninvasive diagnosis of equipment insulation. For the samples in the laboratory, the $Q(t)$ system can simulate the measurement environment within several hundred kilovolts per millimeter and 150 °C temperature. Meanwhile, due to the flexible electrode design, the $Q(t)$ system can be directly connected to the equipment for insulation measurement. At present, the method has been used in the insulations of cables, IGBT modules, and wires in radiation environments [19]–[21], and there are no specific requirements for the temperature and voltage conditions of the equipment during the measurement process. Therefore, the $Q(t)$ method is also widely used to judge the insulation state.

In the previous joint research using the PEA and $Q(t)$ methods, the validity of the $Q(t)$ method has been verified by the consistency of the two methods in indicating the charge accumulation of insulating materials [22], [23]. However, the difference between the charge results of the

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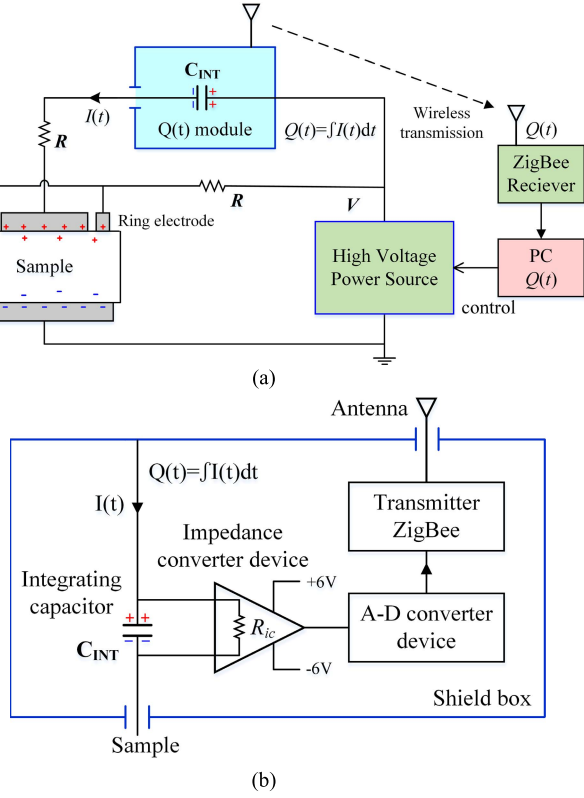


Fig. 1. Measurement system of the $Q(t)$ method. (a) Circuit of the $Q(t)$ method. (b) Internal structure of the $Q(t)$ module.

methods has not been fully discussed. In addition, based on the PEA method, a relatively complete calculation algorithm of distorted electric field and related characteristics has been established. However, in the research of the $Q(t)$ method, the ratio of the measured charge amount to the initial induced amount is only used to evaluate the state of accumulated charge [24]. The involved insulation properties and parameter calculations have not been well-analyzed.

Under this background, we further studied the charge characteristic of each part of the $Q(t)$ data based on the principles of the PEA and $Q(t)$ methods. The evaluation of the insulation properties under the joint application of the two methods is also discussed. The second section of this article derives the relationship between the measured charge ratio and the dielectric relaxation time. The algorithm of calculating the material conductivity is also given. The third section compares the charge characteristics of different materials measured by the two methods, focusing on the analysis of the difference between the two methods and the necessity of joint application. The related results can provide an analysis method for the insulation property and state.

II. BASIC PRINCIPLES OF THE PEA METHOD AND THE $Q(t)$ METHOD

A. Measurement Circuits of the Two Methods

The measurement system of the $Q(t)$ method is shown in Fig. 1. The measured sample is inside a three-electrode module, including a high-voltage electrode, a ring electrode, and a grounded one. The current passing through the high-

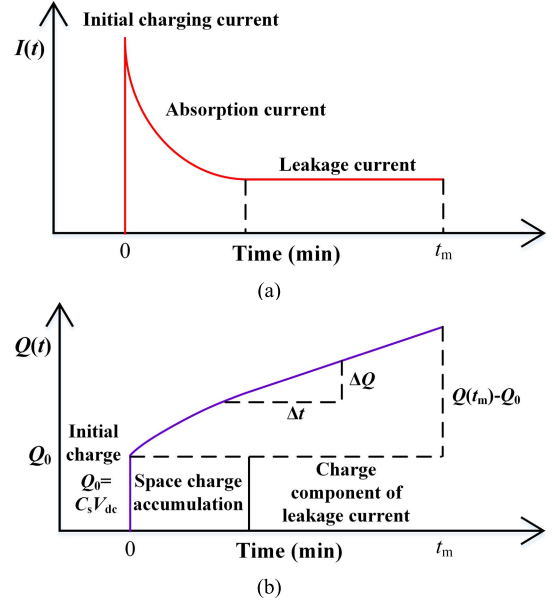


Fig. 2. Three charge components measured by the $Q(t)$ method under an applied dc voltage. (a) Current components. (b) Corresponding charge components.

voltage electrode is the same as that through the integration capacitor C_{INT} , and thus, the charge characteristic can be evaluated. The ring electrode is used to decrease the effect of leakage current from the sample surface.

The minimum time resolution of the actually established system is 10 ms. When the measurement duration is more than 1 day, the resolution is usually set to 1 second. In the aspect of sensitivity, the minimum voltage sensitivity of the A-D converter is about 10 mV. When using an integration capacitor of 0.1 μF , the minimum resolvable charge value is about 10^{-9} C. The measurement process is based on the computer control and wireless signal transmission. After the measurement conditions are input into the computer software, a high dc voltage is applied to the sample. The measured signals are converted by the $Q(t)$ module in Fig. 1(b) and then transmitted to the ZigBee receiver, which are finally recorded by the computer. The stability of the system has been verified based on a long-time leakage current measurement of the $Q(t)$ module.

Under an applied dc voltage, the current passing through the sample mainly contains three components, as shown in Fig. 2(a). In the figure, the time zero means the high voltage is just applied to the sample, and t_m indicates the actual measurement time. The $Q(t)$ method measures the charge information from the beginning of the application of high voltage, and thus, the obtained charge results contain all the charge information including the initial induced charge and the polarization charge part. After the initial charging current reaches the maximum value quickly, the absorption current generally dominates within the dielectric relaxation time. After it approaches zero, the constant leakage current dominates. At this stage, the current and electric field states inside the material are basically stable, which is therefore a proper time to measure the steady dielectric parameters. This current process represents the changing process from

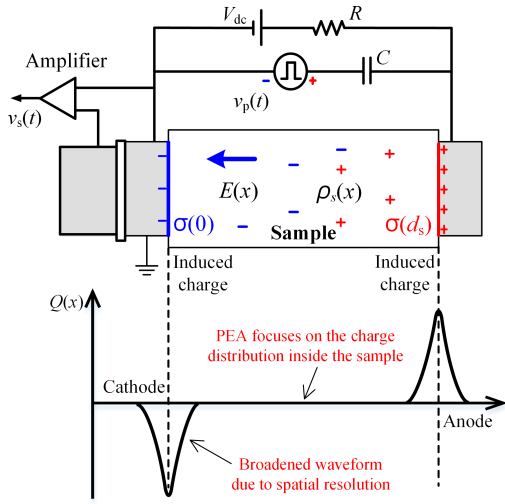


Fig. 3. Measurement circuit of the PEA system.

the capacitive current to the resistive one under an applied high voltage. In the actual measurement, the measurement time t_m depends on the insulating property of the selected sample, which is usually in the range from 5 minutes to several hours. The longest measurement time for some samples reaches several days.

Corresponding to the three current components, the integrating charge results are shown in Fig. 2(b). The dominant processes of the three charge components are similar to those of the current. The initial part is the surface-induced charge under an applied high voltage. Then, space charge injects and accumulates in the sample. There are usually shallow and deep traps inside the sample. The space charge captured by deep traps is difficult to migrate, while the charge in shallow traps can easily migrate toward the opposite electrode. During the injection, migration, and accumulation of space charge inside the sample, it can directly affect the electric field strengths at the two sample surfaces. Thus, it can also change the amount of surface-induced charge, and correspondingly absorption current appears. The whole process must satisfy Kirchhoff's law. When the space charge escapes from the sample, it forms the leakage current. In fact, the process of charge accumulation and migration in deep and shallow traps is the transition from the capacitive current to the resistive one of the dielectric. The $Q(t)$ method can accurately measure these three charge information, that is, it can obtain the changing process of the charge amount integrated by the current over time in Fig. 2(a). In addition, if the sample has some residual charge, the $Q(t)$ system can be directly connected to the sample without applying a high voltage. Then, the residual charge gradually migrates out of the sample, which can be measured by the $Q(t)$ system. Thus, the $Q(t)$ method can also be used in depolarization conditions. However, at present, we mainly use the measured charge amount during a dc voltage application to judge the insulating state of the sample without residual charge.

The basic circuit of the PEA method is shown in Fig. 3. After applying a high voltage to the sample, surface charges induce the sample and space charge is injected into it.

The pulse voltages are applied to the sample to disturb the charge layer. The pressure waves representing the accumulated charge then generate and transmit to the piezoelectric sensor. Finally, the signals are amplified and recorded by the computer.

When a dc voltage is applied to the PEA system, surface charges immediately induce the two interfaces formed by the sample and the electrodes. The width of the surface-induced charge is very narrow, as shown by $\sigma(0)$ and $\sigma(d_s)$ in the upper part of Fig. 3. However, affected by the capacitance of the transducer and the input impedance of the amplifier in the PEA system, the actually obtained surface charge waveform is always close to the Gaussian shape, whose pulsedwidth is the spatial resolution of the PEA system. Therefore, for the sample with space charge inside it, the PEA result shows the net charge distribution at each position along the thickness direction under the effect of spatial resolution. In contrast, the $Q(t)$ method shows the charge amount caused by the net current integration flowing through the sample. Thus, the net charges shown by PEA and $Q(t)$ methods are different concepts. The $Q(t)$ method can measure the total charge amount with time, which can further calculate the three current components and related dielectric properties. In contrast, the PEA method mainly shows the net space charge distribution along the thickness direction, which can be used to analyze the distorted electric field. Therefore, the two methods can be used to analyze the different properties of the material.

B. Measured Charge Components of the $Q(t)$ Method

From Fig. 2, the $Q(t)$ result is composed of initial induced charge Q_0 , space charge component $Q_{\text{spac}}(t)$, and leakage charge $Q_{\text{leak}}(t)$, which can be represented by the following equation:

$$Q(t) = Q_0 + Q_{\text{spac}}(t) + Q_{\text{leak}}(t) \\ = Q_0 + \int_0^t I_{\text{absor}}(t)dt + \int_0^t I_{\text{leak}}(t)dt \quad (1)$$

where Q_0 is the product of the sample capacitance C_s and the applied voltage V_{dc} . $I_{\text{absor}}(t)$ and $I_{\text{leak}}(t)$ are the absorption current and leakage current, respectively.

Since the initial charging process is rapid, the amount of induced charge can be obtained at the beginning of the measurement. The charge ratio $Q(t)/Q_0$ is usually used to evaluate the charge characteristic of insulating materials, as shown in the following equation:

$$\frac{Q(t_m)}{Q_0} = 1 + \frac{Q_{\text{spac}}(t_m)}{Q_0} + \frac{Q_{\text{leak}}(t_m)}{Q_0}. \quad (2)$$

Therefore, according to the above equation, the space charge injection can be preliminarily judged based on whether the charge ratio is larger than 1. Since the initial induced charge can reach the peak value soon after the high voltage is applied, it is easy to obtain the value Q_0 at the initial stage of the measurement. The proportional equation used here can facilitate the derivation and analysis of subsequent equations.

C. Derivation of the Relationship Between the Charge Ratio, Dielectric Relaxation Time, and Measurement Time in $Q(t)$ Data

The initial induced charge amount can be easily measured. Then, the dominating process of the other two charge components is analyzed. The following approximate situation is given.

Under an applied high voltage, the charge with a density ρ is injected into the sample and then begins to migrate. Before it reaches the opposite electrode, there is nearly no leakage current. This is based on the assumption that there are few impurities in the sample and the ionization of them cannot generate a large amount of hetero-charges. Meanwhile, it is also thought that in the early stage of measurement, little space charge injected from one electrode extracts from the other electrode and becomes the leakage current. The migration time T_{tran} of charge can be approximately calculated by the following equation:

$$T_{\text{tran}} = \frac{d_s}{v_c} = \frac{d_s}{\mu_c E} \quad (3)$$

where d_s is the sample thickness, v_c is the drift velocity of charge, E is the applied electric field, and μ_c is the charge mobility. In fact, if space charge accumulates inside the sample, both the electric field and the charge mobility will change. The symbols used here can be thought as the average electric field and charge mobility. According to the measured results in [4], [14], [25], the distribution of space charge can greatly affect the field strengths at different positions inside the sample. However, since the behavior of internal space charge can also change the induced charge on the surfaces, the average electric field of the entire sample is still close to the average value caused by the applied high voltage.

The electrical conductivity k can be used to calculate the charge mobility, that is, $\mu = k/\rho$. It is an approximate equation to calculate the value. Then, the following equation can be derived from (3):

$$T_{\text{tran}} = \frac{\varepsilon}{k} \times \frac{\rho \times d_s \times S}{\sigma \times S} = \tau \times \frac{Q_{\text{spac}}}{Q_0} \quad (4)$$

where the density σ of induced surface charge is equal to the product of permittivity ε and electric field. $\tau = \varepsilon/k$ is the dielectric relaxation time. $Q_0 = \sigma S$ represents the induced surface charge. S is the effective area during measurement. $Q_{\text{spac}} = \rho d_s S$ is the amount of space charge component migrating inside the sample that has not yet become the leakage current. ρ is the average charge density inside the sample.

From (4), when the transit time T_{tran} is equal to the relaxation time τ , it can be found that the value Q_{spac} is equal to Q_0 . At this time, the charge just reaches the opposite electrode, and then the leakage current appears. Therefore, the total charge amount is nearly equal to twice Q_0 according to (1). Here, the relaxation time refers to the time required for the current flowing through the sample to reach equilibrium after a high voltage is applied, at which state the leakage current is the constant conduction current. Before the measurement time

reaches the dielectric relaxation time, the space charge formation current is basically the dominant one in the measured results.

Moreover, the conduction current flowing through the sample can be represented by (5). According to the previous measured results in [21], [26], $Q(t_m)$ can be regarded as a linear upward trend with time based on the induced charge amount Q_0 . Therefore, current is approximated as the ratio of $Q(t_m) - Q_0$ to measurement time t_m here. Meanwhile, the electric current in the steady state is also expressed by the second row of (5), which is eventually represented by the ratio of Q_0 and τ . Here, the dielectric relaxation time is the product of resistance R_s and electrostatic capacitance C_s . From the basic relationship that $R_s = d_s/(k \cdot S)$ and $C_s = \varepsilon \cdot S/d_s$, the relationship of relaxation time and conductivity can also be obtained, that is, $\tau = R_s \cdot C_s = \varepsilon/k$.

$$\begin{cases} I = \frac{\Delta Q(t)}{\Delta t} \cong \frac{Q(t_m) - Q_0}{t_m} \\ I = \frac{V}{R_s} = \frac{Q_0/C_s}{R_s} = \frac{Q_0}{R_s C_s} = \frac{Q_0}{\tau} \end{cases} \quad (5)$$

It should be noted that based on the approximation described above, the equation in the first row of (5) is an approximate solution, whose value is considered to be close to the current in the second row of (5). The following equation can then be derived from (5):

$$\frac{Q(t_m)}{Q_0} = 1 + \frac{t_m}{\tau} \quad (6)$$

This equation shows the mutual relation of the measured charge ratio $Q(t_m)/Q_0$, measurement time t_m , and dielectric relaxation time τ . When $t_m = \tau$, the ratio is equal to 2, and the value of Q_{spac} is equal to Q_0 based on the condition $Q_{\text{leak}} = 0$. In fact, based on (2), it can be inferred that when $Q_{\text{leak}} = 0$ is assumed, the charge is injected into the sample to form Q_{spac} , whose amount gradually increases with time. However, the time point when Q_{spac} is equal to Q_0 cannot be directly calculated from (2). Meanwhile, it is also difficult to know the application range of $Q_{\text{leak}} = 0$ without the above derivation process. The derived equation here also shows the relationship between the measured charge ratio and the relaxation time. At this time, the absorption current due to the space charge accumulation inside the sample basically reaches the steady state. Then, the leakage current from the shallow trapped charge appears and dominates in the later measurement process. It also means that when the charge ratio is smaller than 2, the component of space charge accumulation inside the sample is dominant. Since the leakage current is constant after reaching the steady state, the real conductivity under the stable state can also be calculated. In addition, the distorted electric field inside the material can be evaluated by the PEA method at this moment.

III. CHARGE CHARACTERISTICS BASED ON THE PEA METHOD AND THE $Q(t)$ METHOD

A. Charge Characteristics of Different Insulating Materials Preliminarily Analyzed by the Two Methods

As stated in the Introduction, the previous researches did not discuss the difference between the PEA and $Q(t)$ methods in

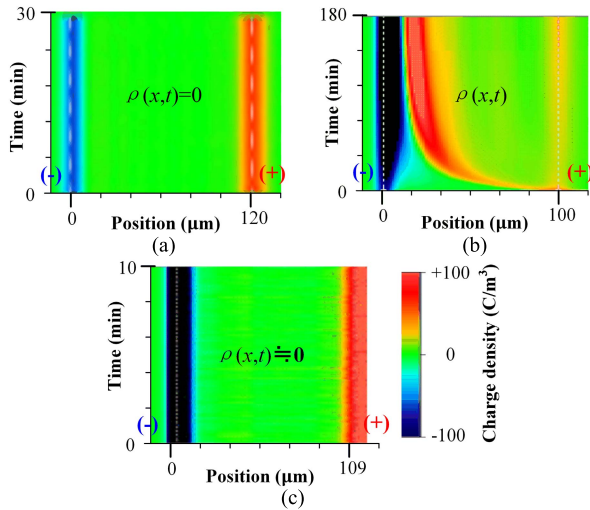


Fig. 4. Charge results of the three materials measured by the PEA method. (a) PC under 100 kV/mm. (b) LDPE under 150 kV/mm. (c) Silicone rubber under 60 kV/mm.

the characterization of charge information. Therefore, we show some typical charge results measured by the two methods here. The measurements are all under positive dc voltages. Fig. 4 shows the PEA results of three samples, including polycarbonate (PC), low-density polyethylene (LDPE), and silicone rubber. The space charge distributions of these materials are measured under different states.

Corresponding to the PEA results, the charge amounts of the same materials under the same conditions are measured by the $Q(t)$ method, as shown in Fig. 5. Therefore, the characteristics based on the two methods can be analyzed.

The PC material shows no space charge accumulation in the two measurements. In the PEA data, only induced surface charges can be found on the two surfaces of the sample. The $Q(t)$ data in Fig. 5(a) are always equal to the initial induced charge amount, which also means there are no charge accumulation and leakage current in the sample. After removing the high voltage applied to the sample at 600 s, the induced charges on the sample surfaces dissipate quickly. Therefore, both the methods show that this PC material keeps a good insulating property under this state.

There is a large amount of space charge in the LDPE material according to the PEA results in Fig. 4(b). Positive charge moves to the cathode and some negative charge is also injected into the sample. The measured $Q(t)$ data are also larger than the initial amount, indicating that there are charge accumulation and leakage current. The first reduction of $Q(t)$ data in Fig. 5(b) is because the high voltage on the sample is removed, and the second one is due to the shutdown of the power supply for the capacitor module. Based on the result in Fig. 4(b), when the system changes to the short-circuit state after the applied dc voltage is removed, a large amount of positive charges are still inside the sample. At this time, the positive charges may be closer to the original cathode side, and most of them still migrate toward the cathode. The direction of current in the short-circuit state is still the same as that in the original voltage application state. Therefore, Fig. 5(b) shows an increase in the $Q(t)$ result under

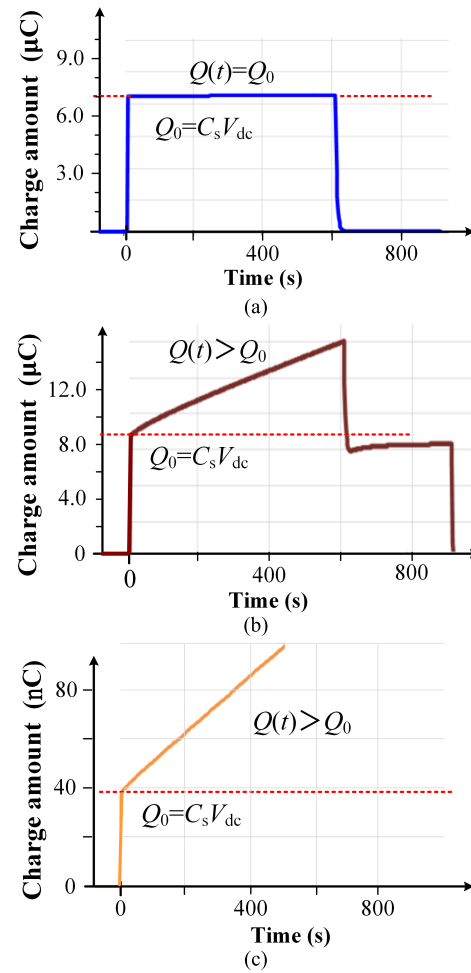


Fig. 5. Charge results of three materials measured by the $Q(t)$ method. (a) PC material. (b) LDPE material. (c) Silicone rubber material.

the short-circuit condition. From the above charge results, this LDPE material cannot keep a good property under this state.

As for the silicone rubber material, the result in Fig. 4(c) shows that there is only a little charge distribution inside the sample, indicating the material may have a good insulating property. However, the $Q(t)$ data in Fig. 5(c) show the opposite phenomenon, indicating that $Q(t) > Q_0$ and the charge accumulation with leakage current appears. It should be noted that the reliability of the $Q(t)$ system has been verified before. The system without any samples shows no leakage current in the long-time measurement.

Therefore, the PEA method can only show the net space charge distribution, which can be used to analyze the related phenomena such as the distorted electric field. Also, according to the founder of the PEA method, it is also difficult to measure shallow trapped charges [15]. In contrast, the $Q(t)$ method can evaluate the total charge information including shallow and deep trapped charges, based on which the leakage current and related phenomena can be discussed. However, the actual charge distribution and distorted electric field cannot be shown. Therefore, it is necessary to use the PEA and $Q(t)$ methods together to correctly evaluate the charge characteristics of insulating materials.

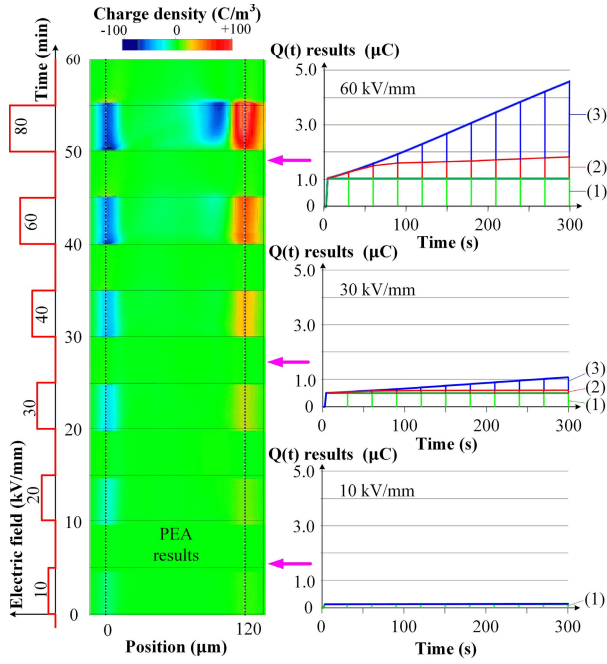


Fig. 6. Charge results of the XLPE samples based on the two methods.

B. Analysis of $Q(t)$ Data Based on the PEA Results

Based on the above analysis, the space charge component Q_{spac} in the $Q(t)$ method is basically composed of all the charge inside the sample which has not become the leakage current. This component dominates before the charge ratio reaches 2 according to the analysis in Section II-3. It is thus different from the net charge information measured by the PEA method. To analyze the proportion of the PEA result in the $Q(t)$ data, the charge density ρ_s measured by the PEA method is used to calculate its total charge amount, as shown in the following equation:

$$Q_{\text{PEA}}(t) = S \int_0^{d_s} \rho_s(x, t) dx. \quad (7)$$

The cross-linked polyethylene (XLPE) films with the thickness of 120 μm are measured. The cross-linking residual byproducts are removed from the samples to clear the impurities. The measured results are shown in Fig. 6. Corresponding to the three electric field strengths of 10, 30, and 60 kV/mm used in the PEA measurement, the three groups of $Q(t)$ data are shown in the right side of the figure. Meanwhile, the measured charge amount by the PEA method is integrated according to (7) and Fig. 7, which is drawn as the red part in the $Q(t)$ data of Fig. 6. The green part is the initial induced surface charge. The blue part represents the charge amount after subtracting the induced part and the PEA result.

From Fig. 6, under an electric field strength of 10 kV/mm, there is nearly no charge injected into the sample. Only the induced surface charges can be found. Therefore, the measured signals using the PEA method under a field strength below 10 kV/mm are usually used as the reference waveforms [27]. When the field strength increases, the charge amounts measured by the PEA and $Q(t)$ methods also increase. Although the field strength only increases by two

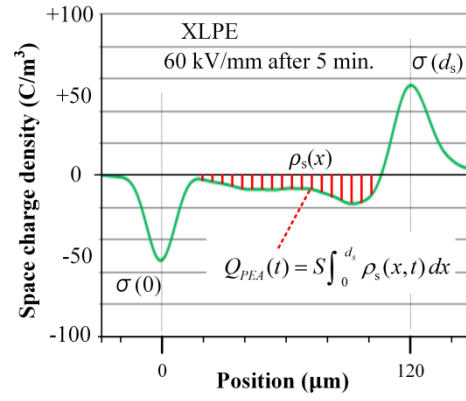


Fig. 7. Integrating part of the PEA result.

times from 30 to 60 kV/mm, the total charge amount increases by four times. Most of the charge proportion is not reflected in the PEA results, indicating that the $Q(t)$ method is a more comprehensive tool to evaluate the whole charge information of insulating materials. Since the PEA method can only measure the net charge distribution, it cannot fully characterize the space charge component Q_{spac} in the $Q(t)$ data. Thus, the time corresponding to the bifurcation of the PEA and $Q(t)$ results in Fig. 6 is not the dielectric relaxation time.

The charge ratios calculated from the results of Fig. 6 are shown in Fig. 8. Moreover, according to (6), Fig. 9 is also given to show the relationship between the charge ratio and dielectric relaxation time. The different curves are the results at different measurement times. Based on the measured charge ratio with its measurement time, the corresponding data point can be determined in the figure. Since the relationship between the relaxation time and conductivity is $\tau = \epsilon/k$, the calculated conductivity is scaled on the upper axis of the figure. The relative permittivity of the used XLPE material is about 2.3.

From Fig. 8, the PEA results can stabilize more easily, while the $Q(t)$ results are always increasing due to the effect of leakage current. The $Q(t)$ result in Fig. 8(a) shows a very small increase, which usually cannot be presented in the PEA result due to the spatial resolution and signal-to-noise ratio of the system. The corresponding relaxation time of the material at this state is very large and the conductivity is very small. The two values cannot be calculated from Fig. 9.

As analyzed in Section II-3, the current basically reaches the steady state when the measurement time is close to the relaxation time, at which the charge ratio is about 2. The final ratio in Fig. 8(b) is close to this value, and thus the material conductivity can be calculated from Fig. 9. It is about 7.2×10^{-14} S/m. Based on the research results in [28], [29], the preparation process of XLPE material can affect its conductivity. The conductivity of the used XLPE material in our laboratory is also measured, whose value ranges from 10^{-13} to 10^{-14} S/m at the applied electric field of 30 kV/mm. The PEA data reach the maximum value at about 100 s, whose ratio is only about 1.2. Since the space charge component dominates in the $Q(t)$ data when the ratio is smaller than 2, it also indicates that the PEA method can only show a small proportion of the space charge information inside the sample.

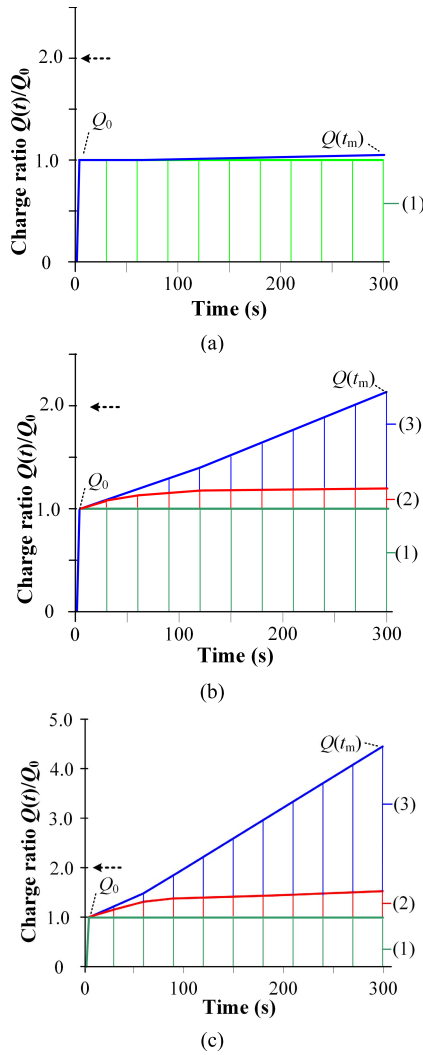


Fig. 8. Charge ratios of XLPE material under different field strengths. (a) Under 10 kV/mm. (b) Under 30 kV/mm. (c) Under 60 kV/mm.

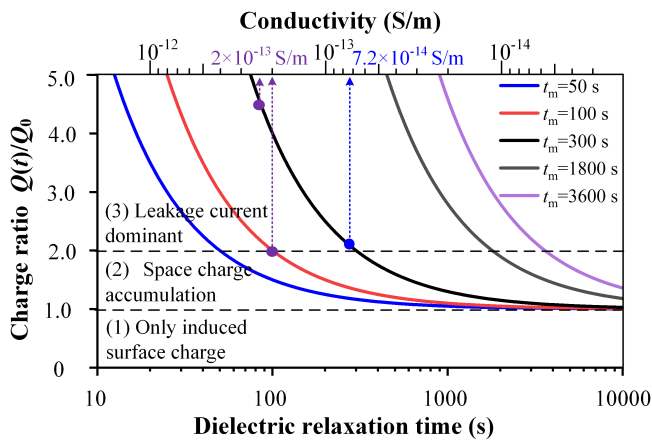


Fig. 9. Calculated relaxation time and conductivity using the $Q(t)$ method.

In Fig. 8(c), the charge ratio is nearly equal to 2 at 100 s. The corresponding conductivity is about 2×10^{-13} S/m from Fig. 9. At 300 s, the final charge ratio is close to 4.5 and the conductivity is about 2.2×10^{-13} S/m, which is very close to

the result at the charge ratio of 2. At this stage, the absorption current is about zero and the leakage current in the material reaches the steady state. Therefore, the $Q(t)$ method can be used to calculate the leakage current according to (1), at which stage it can also evaluate the dielectric relaxation time and the real conductivity by (6) using the measured charge ratio. In addition, the ratio of the PEA part exceeds 1.5 at 500 s, which is closer to 2 compared with the result under the field strength of 30 kV/mm. It indicates that the charge amount measured by the PEA method is closer to the space charge component in the $Q(t)$ data under a higher electric field.

It should be noted that the method using the $Q(t)$ data to calculate the conductivity is based on some assumptions, which have been described in detail when deriving the equations above. Thus, compared with the methods that directly measure the conductivity after a long-term polarization time, the calculated results of the $Q(t)$ method still have errors. However, the $Q(t)$ method can be directly applied to the insulation measurement of cables, wires, and equipment modules due to its high signal-to-noise ratio and flexible electrode designs. The improvement of the theoretical method and practical application of the $Q(t)$ method in the calculation of conductivity and other parameters will be the direction and focus of further research. In addition, the method derived above is for the sample without residual charge before the measurement. For the sample with residual charge, the same type of sample without charge can be measured first to obtain the initial induced charge amount. Then, we can measure the sample with residual charge using the $Q(t)$ method, and the derived method above can be used for further calculation and judgment.

IV. CONCLUSION

The evaluation method of the charge characteristics of insulating materials based on the PEA and $Q(t)$ methods is discussed and studied in detail. By an equation derivation and experiment analysis process, three main conclusions are obtained.

- 1) The relationship between the ratio of the measured charge amount to the initial induced amount, dielectric relaxation time, and measurement time is derived. Preliminary analysis finds that when the measured charge ratio is larger than 2, the absorption current of the insulating material tends to zero. Meanwhile, the leakage current reaches the steady state, and thus, the current and electric field become stable.
- 2) Based on the measured results of different materials using the PEA and $Q(t)$ methods, it is pointed out that the PEA result can only reflect the net charge distribution captured by deep traps inside the sample, and thus, it is usually used to calculate the distorted electric field. In contrast, the $Q(t)$ method can get the changing process of all charges induced and injected in the material, which is applicable to analyze the leakage current. Therefore, only the combination of the two methods can comprehensively and effectively evaluate the charge characteristics of dielectric materials.

- 3) Using the derived relationship between the charge ratio and relaxation time in the $Q(t)$ method, the experimental results further indicate that when the measured charge ratio is close to 2, the leakage current, dielectric relaxation time, and conductivity of the insulating material can be calculated. Meanwhile, the value of conductivity no longer changes significantly even after the ratio is much larger than 2. Therefore, the $Q(t)$ method provides a new way for the calculation and evaluation of electric charge, current, and related characteristics under high-voltage dc conditions.

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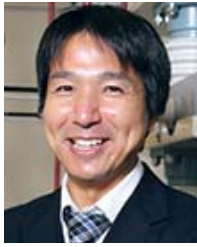
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